

(325) Network Design Team

Description & Eligibility

Analyze existing and planned business environments and develop a strategy for the implementation of a network infrastructure that addresses the business needs of the scenario provided. Any postsecondary or secondary division contestant may enter this judged event. A team will consist of 2-4 contestants. Teams must be registered for the national competition before the submission deadline and compete in both parts of the event to be ranked. **Entries that do not follow the provided topic or the Copyright and/or Fair Use Guidelines will be disqualified.**

Topic

Voyager Virtual Reality Ventures, headquartered in Seattle, Washington, operates three pop-up VR arcade kiosks inside major shopping malls. Each kiosk features four high-performance VR pods, a spectator lounge with digital signage, and a small retail counter selling VR accessories. To date, each location has relied on disparate cellular hotspots and aging consumer-grade routers, resulting in inconsistent performance, high latency during peak hours, and unsecured guest networks.

As part of their growth strategy, Voyager plans to open three full-service VR Experience Centers in 2026—one in Austin, Texas; Raleigh, North Carolina; and Portland, Oregon. These new locations will be permanent retail spaces of approximately 3,500 square feet, with dedicated server rooms, wired and wireless network zones for VR pods, administrative offices, a streamer's booth for live E-Sports events, and a café area offering complimentary guest Wi-Fi.

Voyager Virtual Reality Ventures needs a proposal for the following:

- High-Throughput, Low-Latency Network Architecture: Design a network capable of supporting simultaneous VR streams (minimum 10 Gbps backbone) with quality-of-service (QoS) to prioritize VR traffic.
- Network Segmentation: Create VLANs to isolate VR pods, corporate offices, guest Wi-Fi, digital signage, and IoT sensors (environmental monitoring, motion tracking).
- Centralized Management and Monitoring: Propose a unified network management solution (e.g., SD-WAN controller or cloud-managed switches/APs) for all six sites, with real-time performance dashboards and automated alerting.
- Security and Remote Support: Incorporate firewalls, intrusion detection/prevention, VPN access for remote troubleshooting, and a plan for role-based access control.
- Scalability and Redundancy: Ensure the design can accommodate at least double the number of VR pods and add-on mobile “VR-on-wheels” kiosks; include redundant uplinks and failover strategies.
- IoT and Digital Signage Integration: Plan for PoE-powered environmental sensors in VR rooms and digital signage in the lounge and café areas, with secure segmentation.
- Cost Estimation and ROI Analysis: Provide a high-level bill of materials (switches, routers, APs, cabling, racks) and estimate both upfront and ongoing costs, balanced against projected revenue gains from improved uptime and customer satisfaction.

Teams should deliver a written proposal in Report Format, complete with network diagrams, equipment specifications, and Works Cited, formatted according to the *Style & Reference Manual*.

Team Must Supply

- Published and/or unpublished non-electronic written reference materials
- Laptop/notebook computer (**each** team contestant may have **one** laptop), portable printer, software, paper and supporting equipment (extension cords, etc.)
- Optional items
 - Digital presentation tools and supporting equipment (device, software, projector, extension cords, etc.).
 - Each team member may have one laptop.

Competition Notes

- Method of evaluation: judged.
- Carry-in and set-up of equipment must be done solely by the team and must take place within the allotted time for orientation/warm-up.
- No Internet access will be provided on-site at NLC; however, contestants may provide their own access to be used only for their presentation to the judges.

- Business Professionals of America assumes no responsibility for hardware/software provided by the Contestant.
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area.
- No previous Business Professionals of America tests (handwritten, photocopied, or keyed) may be taken into the testing area. **Violation of this rule will result in disqualification.**
- Length of event: no more than three (3) minutes setup in presentation room, no more than ten (10) minutes presentation time, and no more than ten (10) minutes of judges' questions.
 - No more than thirty (30) minutes preparation time at state and national level only
 - Teams will be presented with an additional element to the scenario that requires revision of their final presentation.
- Finals may be part of the competition.
- Each state is allowed three (3) entries.

Technical Pre-submit Specifications

What to Submit at https://upload.bpa.org	Saved File Naming Convention	National Deadline
Written Proposal, Works Cited and BPA Release Form(s) in one combined PDF file.	NDT-TeamID.pdf	April 1, 2026, 11:59 p.m. Eastern Time

Contest Competencies

- Apply technical skills in network design
- Evaluate and delegate responsibilities needed to perform required tasks
- Utilize problem solving techniques
- Demonstrate teamwork skills needed to function in a business setting
- Analyze existing and planned network and systems management
- Demonstrate knowledge of network cable topologies
- Diagram the completed network
- Analyze end-user work needs
- Plan for placement and management of resources
- Plan for decentralized resources or centralized resources
- Incorporate redundancy into the network
- Design a remote access solution
- Analyze network infrastructure, protocols, and hosts
- Estimate upgrade cost based on existing environment
- Incorporate future expansion into the network
- Demonstrate an ability to meet software requirements
- Demonstrate a knowledge of VPNs
- Identify the company's tolerance for risk
- Analyze technical support structure
- Design a routing strategy
- Develop multiple recommendations to scenarios
- Produce business reports

Contest Specifications for this Event

- The written proposal must follow the report format located in the *Style & Reference Manual*.
- At the state and national levels, teams will be given an additional element to the original scenario to address during the presentation. These added elements are specific to the level of competition and are not connected to or carried over from previous competitions.
- Cell phones may *not* be used in the preparation room.
- No advisor contact will be allowed between the time of receiving the additional element and the delivery of the presentation.
- Only team-prepared graphics may be used in the design. Originality and effectiveness will be given greater consideration.

Common Specifications for all BPA Judged Events

- This is a pre-submitted event. Regional and State Conferences have specific deadlines prior to the National deadline. Please consult the regional or state conference administrator for regional or state deadlines. For the National Leadership Conference, submit the URL to <https://upload.bpa.org> no later than 11:59 p.m. Eastern Time, on April 1, 2026.
- Team ID is required for all submissions. Materials from non-registered contestants and/or those missing Team ID cannot be accepted.
- The team is responsible for securing a signed BPA Release Form(s) from any person, including themselves, whose image or work is used in the project.
- The team is responsible for citing all sources, including themselves, on the Works Cited page for any work used on the project. It is the policy of Business Professionals of America to comply with copyright law. The *Style & Reference Manual* contains guidelines for Copyright and/or Fair Use. Teams will be disqualified for violations of the guidelines.
- Materials must follow the organization's BPA Graphic Standards and make proper use of the BPA logo and/or organization's name. (Refer to the BPA Graphic Standards in the *Style & Reference Manual*).
- Judges' comments will be returned digitally through the online judging system at the national level. Materials submitted for technical judging will *not* be returned and will *not* be available at NLC.

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Technical Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Submission and Formatting (All points or none are awarded by the technical judge)				
Team submitted one (1) combined PDF file containing their Written Proposal, Works Cited formatted according to the <i>Style & Reference Manual</i> and BPA Release Form(s).				/10
Written Proposal	Below Expectations 0-6 points	Meets Expectations 7-13 points	Above Expectations 14-20 points	Points Awarded
Customer Profile	Incomplete or unclear; lacks relevance to topic.	Clear and relevant; basic customer needs identified.	Thorough and insightful; customer needs and context well-defined.	/20
Objectives	Vague or missing; lacks measurable goals.	Clear and measurable; aligned with topic.	Strategic and impactful; objectives show depth and foresight.	/20
Abstract of Implementation	Lacks clarity or detail; hard to follow.	Clear summary of implementation steps.	Exceptionally clear and well-structured; shows strong understanding.	/20
Explanation of Products/Services	Minimal or unclear explanation.	Adequate explanation with relevance to topic.	Detailed and compelling; clearly demonstrates value.	/20
Clarity of Message	Message is unclear or off-topic.	Message is clear, concise, and relevant	Message is compelling, creative, and perfectly aligned with topic.	/20
Short- and Long-Range Goals	Goals are missing or poorly defined	Goals are defined and realistic	Goals are strategic, ambitious, and well-supported.	/20
Financial Analysis	Missing or inaccurate; lacks justification.	Basic financials with reasonable justification.	Detailed, accurate, and insightful financial planning.	/20
Supporting Documentation	Missing or poorly organized.	Includes relevant and organized documentation.	Comprehensive, well-organized, and enhances understanding.	/20
Writing Accuracy	Frequent grammar or spelling errors.	Contains some grammar or spelling errors.	Contains very few or no grammar or spelling errors.	/20
Creativity	Below Expectations 0-6 Points	Meets Expectations 7-13 Points	Above Expectations 14-20 Points	Points Awarded
Innovation of Solution	Conventional or derivative ideas.	Some originality; meets basic expectations.	Highly innovative; fresh, unique, and creative approach.	/20

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Technical Scoring Rubric (Con't)

Specific Technical Recommendations	Below Expectations 0-10 Points	Meets Expectations 11-30 Points	Above Expectations 31-50 Points	Points Awarded
Design & Deployment Specifications	Missing or vague specifications.	Includes basic design and deployment details.	Detailed and technically sound; exceeds expectations.	/50
Specific Technical Recommendations	Below Expectations 0-6 Points	Meets Expectations 7-13 Points	Above Expectations 14-20 Points	Points Awarded
Network Structure & Setup	Incomplete or unclear setup.	Basic structure and setup provided.	Robust and scalable setup; clearly supports requirements.	/20
Bandwidth Tiers & Services	Missing or not well-defined.	Identifies basic tiers and services.	Clearly defined tiers; tailored to customer needs.	/20
Scalability & Quality of Service	No mention of scalability or QoS.	Addresses scalability and reliability.	Strong emphasis on scalability, QoS, and future-proofing.	/20
Value-Added Services	Not identified or irrelevant.	Identifies basic enhancements.	Creative and strategic enhancements that differentiate solution.	/20
TOTAL TECHNICAL POINTS				/340

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Presentation Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines.				☐ Y ☐ N
If answered no, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Evaluation Criteria	Below Expectations 0-5 points	Meets Expectations 6-10 points	Above Expectations 11-15 points	Points Awarded
Delivery & Engagement	Team rarely maintains eye contact, uses minimal gestures and body language, or has significant inconsistencies in tone and pace.	Team occasionally breaks eye contact, uses limited gestures and body language, or has some inconsistencies in tone and pace.	Team maintains good eye contact, uses appropriate gestures and body language, and varies their tone and pace adequately.	/15
Organization, Design & Structure	Presentation lacks coherence, and it is unclear how each team member's role contributes to the presentation. Transitions between speakers and/or points are awkward or absent.	Presentation content has some gaps, and it is not always clear how each team member's role contributes to the presentation. Transitions between speakers and/or points are choppy.	Presentation content flows adequately, and each team member's role is clear and integrated into the presentation. Transitions between speakers and/or points are smooth.	/15
Evaluation Criteria	Below Expectations 0-15 points	Meets Expectation 16-30 points	Above Expectations 31-45 points	Points Awarded
Content Knowledge	Presentation demonstrates a limited understanding of the topic and includes mostly irrelevant or inaccurate information.	Presentation demonstrates a basic understanding of the topic and includes a mix of relevant and irrelevant information.	Presentation demonstrates a good understanding of the topic and includes relevant and accurate information.	/45
Evaluation Criteria	Below Expectations 0-4 points	Meets Expectations 5-7 points	Above Expectations 8-10 points	Points Awarded
Judges' Questions	Responses lack depth and fail to demonstrate a comprehensive understanding of the topic.	Some responses lack detail or clarity. The understanding of the topic is partially demonstrated.	Responses are detailed, insightful, and effectively address the judges' questions.	/10
Team Members	At least two original team members were present during the presentation.			/5
Timing	Setup lasted no longer than three (3) minutes, and the presentation lasted no more than ten (10) minutes.			/10
TOTAL PRESENTATION POINTS				/100

PRESENTATION WILL BE STOPPED AT TEN (10) MINUTES